Decidability 03

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Undecidability

- Consider the following language:

$$A_{\text{TM}} = \{\langle M, w\rangle \mid M \text{ is a TM and } M \text{ accepts } w\}$$

- Is A_{TM} recognizable?
- To show that A_{TM} is recognizable, you need to construct a TM that accepts all strings in A_{TM}
  - But if a string is not in A_{TM}, no need to explicitly reject
  - Loop indefinitely is acceptable
- The following is an example of a TM that recognizes A_{TM}:

  $U =$ "On input $\langle M, w\rangle$, where M is a TM and w is a string:

  1. Simulate M on input w.
  2. If M ever enters its accept state, *accept*; if M enters its reject state, *reject*."

Undecidability

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- Let's look at the above machine:
  - If $\langle M, w\rangle \in A_{\mathrm{TM}}$:
    - M is a TM that accepts w
    - TM U will accept $\langle M, w\rangle$
  - If $\langle M, w\rangle \notin A_{\mathrm{TM}}$:
    1. M is a TM that rejects w
       TM U will reject $\langle M, w\rangle$
    2. M is a TM that loops indefinitely on input w
       TM U will loop indefinitely on input $\langle M, w\rangle$

Undecidability

- TM U accepts all strings in A_{TM} and **does not accept** (either rejects or loop indefinitely) on all strings not in A_{TM}
  - A_{TM} is recognizable
- Is A_{TM} decidable?
  - Unfortunately, A_{TM} is undecidable
  - But how to prove it?
- We will try to prove that it is undecidable later

Unrecognizable Lanugage

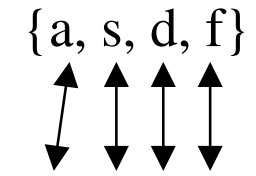
Corollary 4.18

Some languages are not Turing-recognizable.

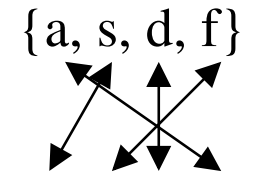
- To show that there exists an unrecognizable language, we only need to find a language that is not recognized by any Turing machines
  - It is not not easy to come up with such a language
- **Note** that a Turing machine recognizes only one language
- To prove that an unrecognizable language exists, we can simply show that **the number of languages is more than the number of Turing machines**
- But we have a problem
  - There are infinite number of languages
  - There are infinite number of Turing machines
- How do we know which set is larger?

Cantor's Observation

- Cantor observed that two **finite** sets A and B have the same size if we can pair one element from A with the element of B



Pairing #1



Pairing #2

- The pairing does not have to be unique
- We can consider pairing method as a function $f : A \to B$ that has the following properties:
  - **one-to-one**: $f(a) \neq f(b)$ whenever $a \neq b$
  - **onto**: For every $b \in B$ there is an $a \in A$ such that $f(a) = b$
- Functions that are one-to-one and onto are called **bijection** or **correspondence**

Example

Same Size?

Let $\mathbb{N}$ be the set of natural numbers $\{1, 2, 3, \dots\}$ and let $\mathcal{E}$ be the set of even natural number $\{2, 4, 6, \dots\}$. Show that $\mathbb{N}$ and $\mathcal{E}$ have the same size.

- To show that $\mathbb{N}$ and $\mathcal{E}$ have the same size, according to the canter's observation, we need to show a correspondence between $\mathbb{N}$ and $\mathcal{E}$
- A mapping function can be $f : \mathbb{N} \to \mathcal{E}$ where $f(n) = 2n$

$n \in \mathbb{N}$	$f(n) \in \mathcal{E}$
1	2
2	4
3	6
4	8
5	10
$\vdots$	$\vdots$

- By Cantor's definition of size, they have the same size

Infinite Ways of Correspondence

- Recall that a correspondence does not have to be unique
- Previous slide show a correspondence between $\mathbb{N}$ and $\mathcal{E}$
- How many correspondence are there between $\mathbb{N}$ and $\mathcal{E}$
  - Infinite

$n \in \mathbb{N}$	$f(n) \in \mathcal{E}$
1	2
2	4
3	6
4	8
5	10
⋮	⋮

$n \in \mathbb{N}$	$f(n) \in \mathcal{E}$
1	4
2	2
3	6
4	8
5	10
⋮	⋮

$n \in \mathbb{N}$	$f(n) \in \mathcal{E}$
1	6
2	2
3	4
4	8
5	10
⋮	⋮

- There are infinite many permutation of $\mathcal{E}$
- Once there is a correspondence between two infinite sets, there are infinitely many correspondences

Countable

Definition 4.14

A set A is **countable** if either it is finite or it has the same size as $\mathbb{N}$.

- To show that an infinite set A is countable, we need to show that it has the same size as $\mathbb{N}$
- To show that an infinite set A has the same size as $\mathbb{N}$, we need to show a correspondence $f : \mathbb{N} \to A$ between A and $\mathbb{N}$
  - Can be a definition of a function (one-to-one and onto)
  - Can be a table that demonstrates one-to-one and onto
- An infinite set B is uncountable if there is no correspondence with $\mathbb{N}$
- An infinite set A is smaller than an infinite set B if A is countable but B is not countable.

The Set of Integers is Countable

- To show that $\mathbb{Z}$ is countable is to show a correspondence between $\mathbb{N}$ and $\mathbb{Z}$
- Example (Invalid Correspondence)

$n \in \mathbb{N}$	$f(n) \in \mathbb{Z}$
1	0
2	1
3	2
4	3
⋮	⋮
?	-1
⋮	⋮

- The above table is not a valid correspondence
  - It is one-to-one
  - It is not onto

The Set of Integers is Countable

- For $\mathbb{Z}$ we need to pair elements from both sides of 0
- Example

$\mathbb{N}$	1	2	3	4	5	6	7	...
$\mathbb{Z}$	0	1	-1	2	-2	3	-3	...

- The above correspondence can be represented by a function $f : \mathbb{N} \to \mathbb{Z}$ where

$$f(1) = 0$$

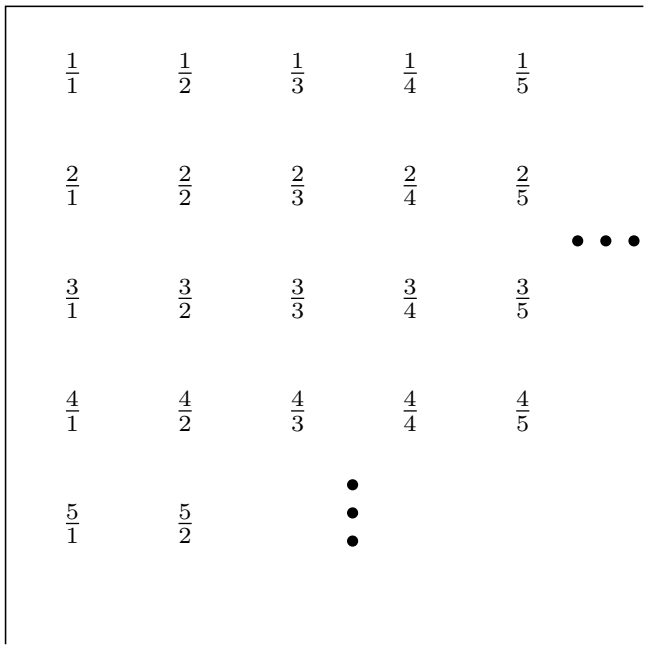
$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is divisible by 2} \\ -\frac{n-1}{2} & \text{if } n \text{ is not divisible by 2} \end{cases}$$

The Set of Positive Rational Numbers is Countable

- Let $\mathbb{Q}$ be a set of rational numbers:

$$\mathbb{Q} = \{\frac{m}{n} \mid m \in \mathbb{N} \text{ and } n \in \mathbb{N}\}$$

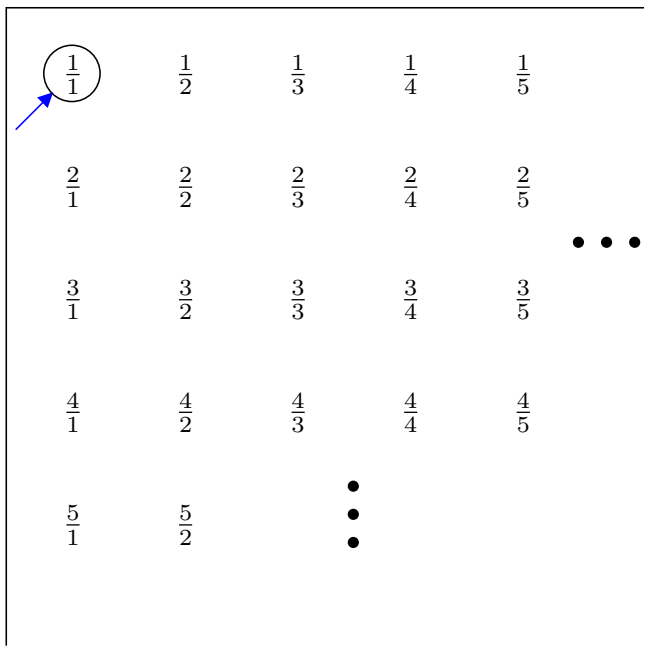
- Again, we need to find a correspondence between $\mathbb{N}$ and $\mathbb{Q}$
- The best starting point is to try to list them all in an order



- Then, try to pair one-by-one but make sure that it will cover all of them

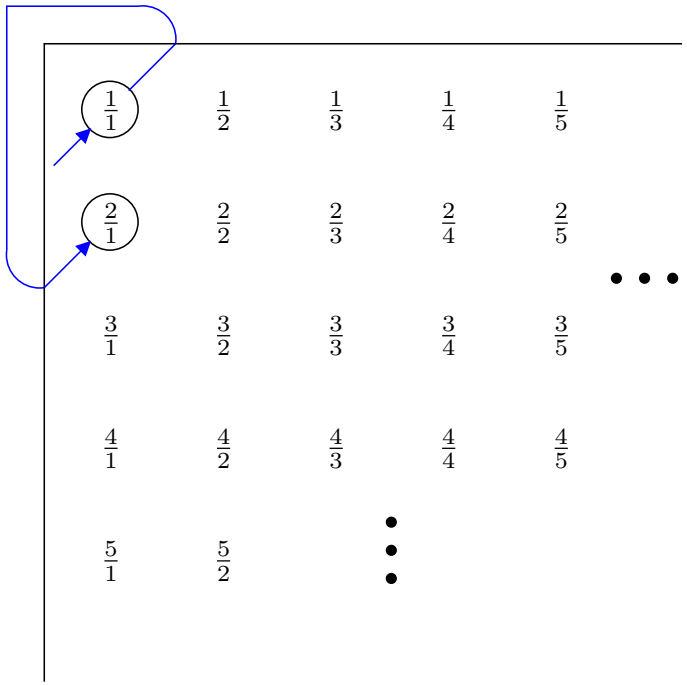
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 1



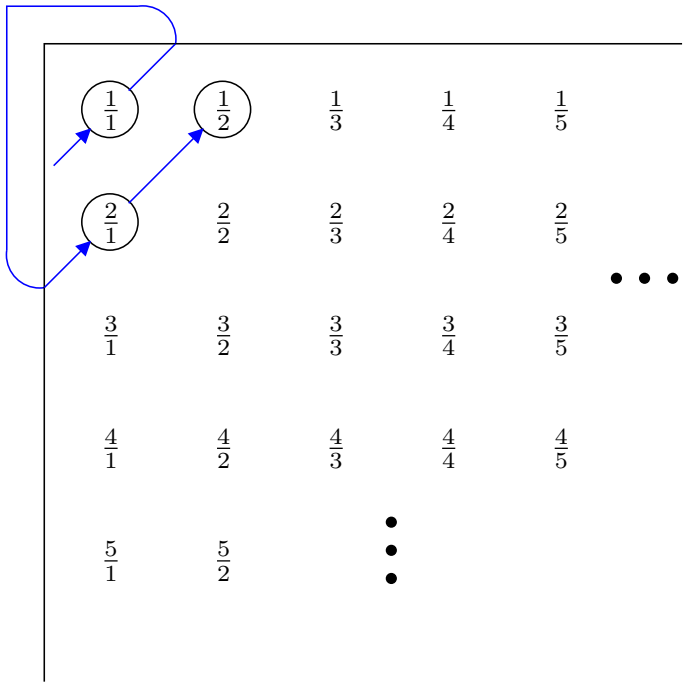
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 2



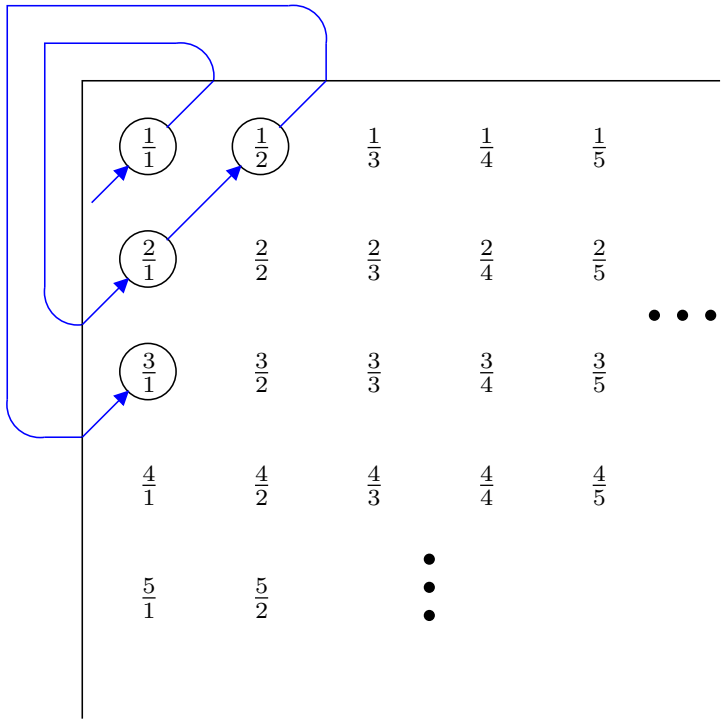
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 3



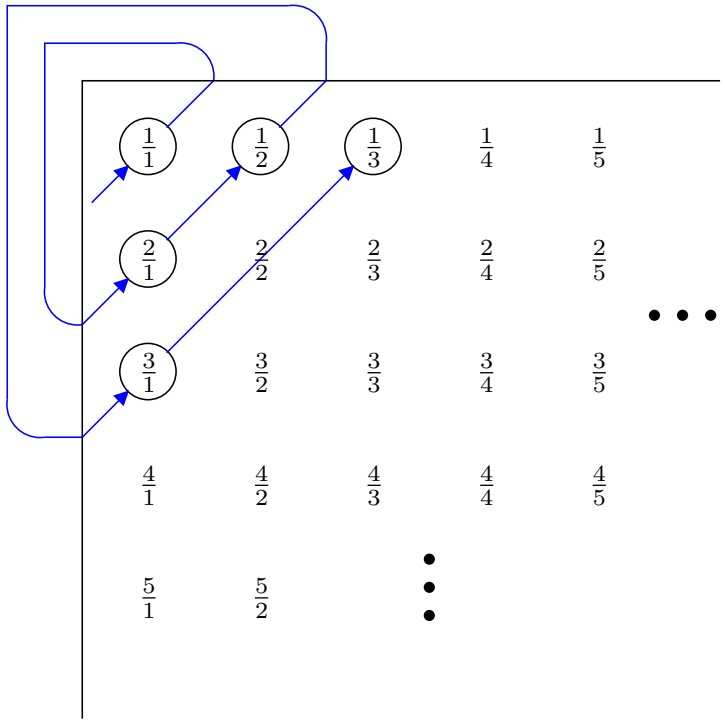
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 4



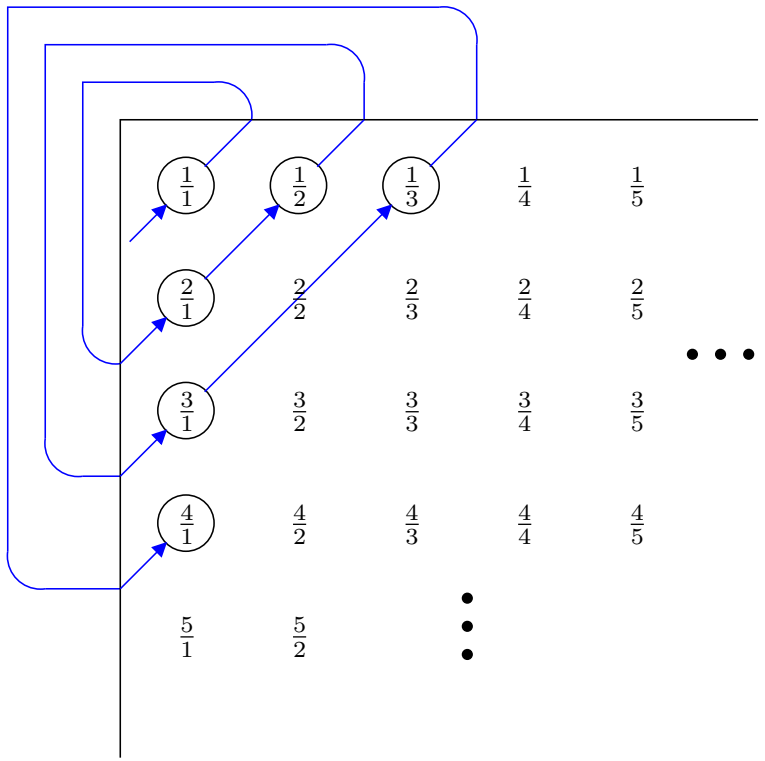
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 5



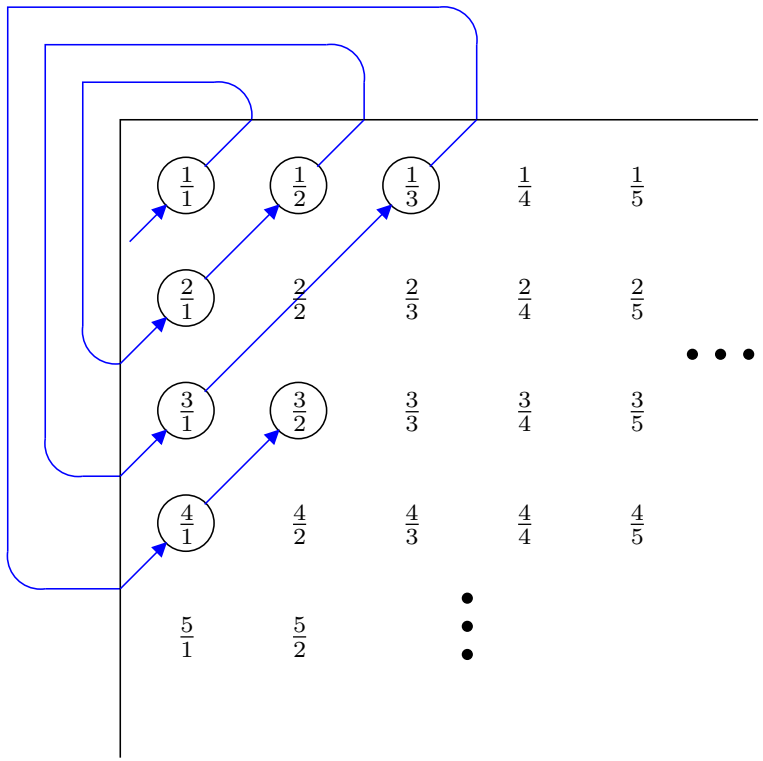
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 6



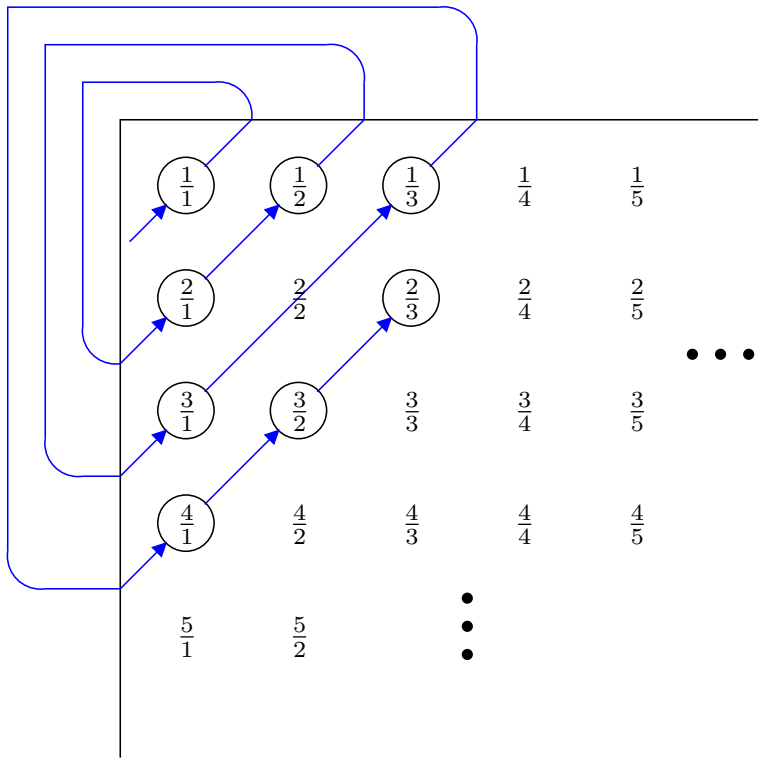
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 7



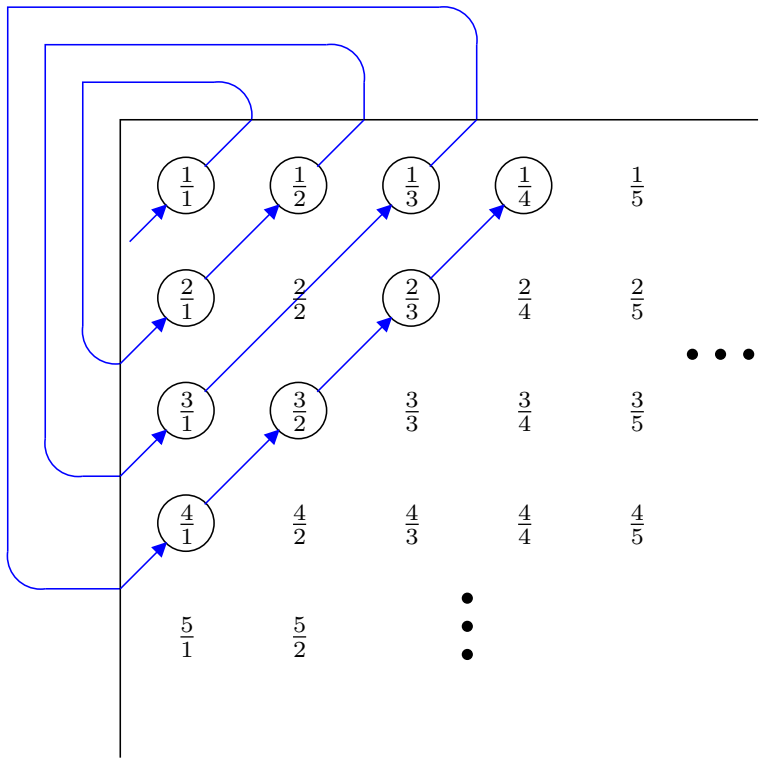
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 8



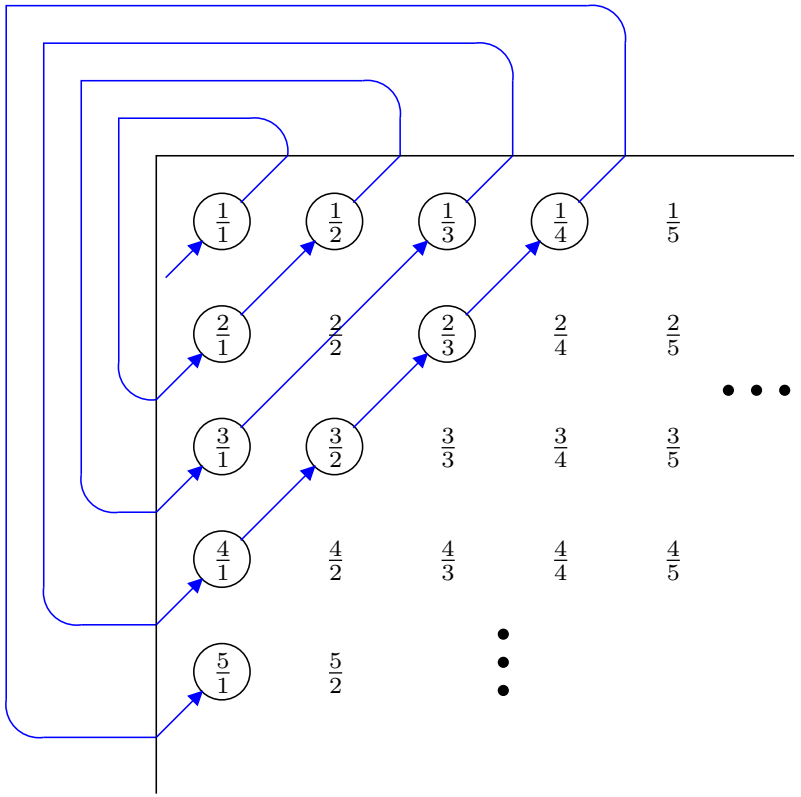
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 9



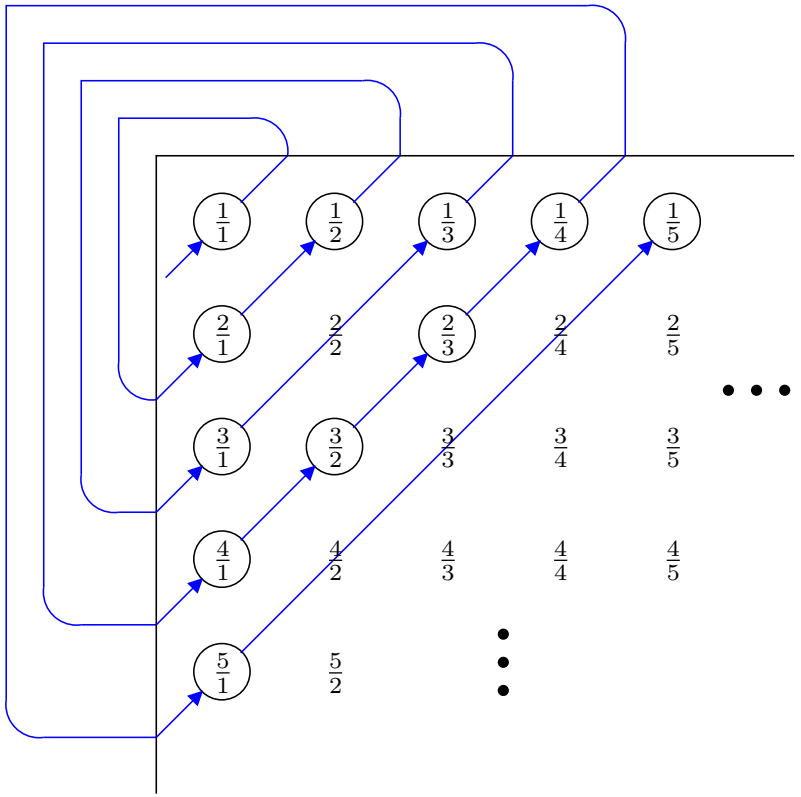
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 10



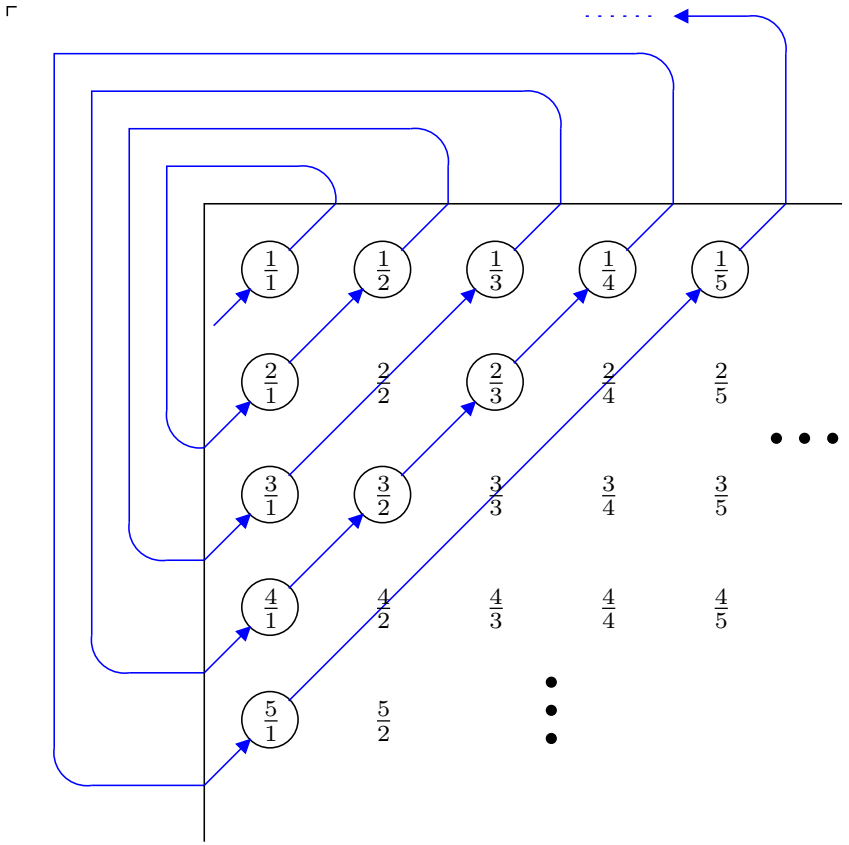
The Set of Positive Rational Numbers is Countable

- Correspond with natural number 11



The Set of Positive Rational Numbers is Countable

- Correspond with natural number greater than 12



- This is an example of a correspondence between $\mathbb{N}$ and $\mathbb{Q}$
- $\mathbb{Q}$ is countable